

Here's the complete Speaking Script for all 12 slides of Session 2: Chunk Enrichment & Vector Databases.

Each script is written in a natural, conversational teaching style with timing estimates.

Slide 1: Title Slide

Duration: ~1.5 mins

"Good morning everyone! Welcome to **Session 2** of our Advanced RAG series — **Chunk Enrichment and Vector Databases**.

In the last session, we learned how to split documents effectively. Today, we'll go one step further: how to make those chunks *intelligent* and where to store them.

We'll cover chunk enrichment techniques and then explore the entire vector database landscape. This session is very important because enrichment and storage decisions heavily impact the quality and scalability of your RAG system.

Any quick questions before we begin? Let's get started."

Slide 2: What is Chunk Enrichment

Duration: ~2.5 mins

"So, what exactly is Chunk Enrichment?

Simply put, it means adding extra useful information to our raw text chunks — like metadata, entities, keywords, or summaries.

Why do we do this? Because raw text chunks are quite basic. Enrichment turns them into smart, searchable units.

Benefits include better retrieval accuracy, powerful filtering, hybrid search, and overall improved RAG performance.

Think of it as upgrading from plain text to richly tagged data."

Slide 3: PII Detection & Redaction

Duration: ~3 mins

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"One of the most critical parts of enrichment is **PII Detection and Redaction**.

PII means Personally Identifiable Information — names, emails, phone numbers, Aadhaar numbers, PAN cards, bank details, etc.

In real-world applications, especially in India, we must follow DPDP Act and GDPR-like regulations.

We can use tools like Microsoft Presidio, regex patterns, or even LLMs to detect and redact sensitive data before storing chunks.

This is not optional in production systems — it's mandatory."

Slide 4: Named Entity Recognition (NER)

Duration: ~2.5 mins

"Next is **Named Entity Recognition (NER)**.

NER automatically extracts important entities from text — such as Person, Organization, Location, Date, Product, etc.

For example, from a news article, it can pull out company names, dates, and locations.

These entities become excellent metadata, which helps in filtering and improves retrieval precision significantly."

Slide 5: Key-Phrase Extraction & Summarization

Duration: ~2.5 mins

"We can also extract **Key Phrases** and create short summaries of each chunk.

Techniques include statistical methods like TF-IDF, RAKE, or modern methods like KeyBERT and LLM-based extraction.

Adding a short summary as metadata helps the retriever understand the chunk's content quickly, even before reading the full text."

Slide 6: Metadata for Hybrid Search

Duration: ~3 mins

"This brings us to one of the most powerful concepts — **Metadata for Hybrid Search**.

When we attach metadata (source, date, category, entities, confidence score, etc.) to chunks, we can do two things:

1. Vector similarity search
2. Metadata filtering

This combination is called Hybrid Search and dramatically improves precision.

Good metadata is often more valuable than using a better embedding model."

Slide 7: Vector Database Landscape

Duration: ~2.5 mins

"Now let's move to the second major topic of today — Vector Databases.

There are many options available today. We can broadly categorize them into:

- Managed services (like Pinecone)
- Self-hosted solutions
- Lightweight in-process databases (like Chroma)
- Databases already in your stack (like pgvector)

We'll explore all of them in detail."

Slide 8: Decision Matrix

Duration: ~3 mins

"Here is a practical Decision Matrix to help you choose the right vector database.

(Go through the table)

For quick development and local testing → Chroma is excellent.

For production with scale → Pinecone or Weaviate.

If you already use PostgreSQL → pgvector is a smart choice."

Slide 9: HNSW vs IVF Indexes

Duration: ~2.5 mins

"Behind the scenes, vector databases use different indexing algorithms.

The two most important ones are **HNSW** and **IVF**.

HNSW gives faster search and better accuracy but uses more memory.

IVF is more memory efficient and better for extremely large datasets.

Understanding this tradeoff helps you make better architecture decisions."

Slide 10: When to Choose What

Duration: ~3 mins

"So, when should you choose which database?

- Starting out or doing local development? → Use **Chroma**
- Already on Postgres? → Go with **pgvector**
- Need maximum scale and ease? → **Pinecone**
- Want full control and open source? → **Weaviate** or **Qdrant**

We'll also see this in the live coding session."

Slide 11: Best Practices

Duration: ~2.5 mins

"Here are some important best practices:

- Always enrich chunks with metadata
- Redact PII before storing
- Keep metadata clean and lightweight
- Test retrieval quality after enrichment
- Balance enrichment cost with performance gains

Enrichment should be thoughtful — not everything needs to be extracted."

Slide 12: Summary & Key Takeaways

Duration: ~2 mins

"To summarize Session 2:

Key Takeaways:

- Raw chunks are not enough — Enrichment adds intelligence
- Metadata turns simple vector search into powerful hybrid search
- Choose your Vector DB carefully based on scale, cost, and control
- Start simple with Chroma, then scale as needed

Remember: **Smart Chunks + Smart Storage = High-Quality RAG**

Any questions on these concepts before we move to the live coding?"

Total PPT Duration: Approximately 28–32 minutes.

Would you like me to now provide:

1. **Detailed Student Notes** for Session 2, or
2. **Live Coding Script** for Session 2?

Let me know!